

MS Research Project Proposal: An AI-Augmented Unified Compliance Auditor for Hybrid Cloud Environments: A Case Study for Rwanda's Regulatory Landscape

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1 Project Objectives

Enterprises and government agencies across Africa are undergoing rapid digital transformation, increasingly adopting hybrid cloud infrastructures. However, compliance and cybersecurity assurance remain fragmented and heavily manual, consuming up to 40% of IT security budgets [1], often requiring more than 1,000 hours annually [2], and introducing error-prone, siloed approaches [1].

At a continental level, the **African Union's Malabo Convention** aims to create a unified legal framework for cybersecurity [3]. However, its implementation has been slow, as the convention is **not yet in force** due to an insufficient number of ratifications. This has created a regulatory gap where individual nations must take the lead. **Rwanda stands out as a proactive leader in this regard**, having established its own comprehensive **NCSA Minimum Cybersecurity Standards in 2023** [4]. This project directly addresses this real-world landscape by focusing on Rwanda's enforceable local standards, providing a timely and necessary solution where continental frameworks are still aspirational.

This project proposes the design and development of an AI-Augmented Unified Compliance Auditor that leverages Natural Language Processing (NLP), anomaly detection, and risk scoring models to automate compliance auditing for hybrid cloud environments. This project will specifically focus on **Rwanda as a pioneering case study**, with measurable objectives to achieve $> 93\%$ compliance-classification accuracy [5,6], reduce audit cycle time by $\geq 50\%$ [7], and ensure explainability in model outputs [8].

1.1 Mathematical Framework for Compliance Classification

The compliance classification accuracy can be formulated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} > 0.93 \quad (1)$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

The audit cycle time reduction is measured as:

$$\text{Time Reduction} = \frac{T_{\text{manual}} - T_{\text{automated}}}{T_{\text{manual}}} \geq 0.5 \quad (2)$$

where T_{manual} is the time for manual auditing and $T_{\text{automated}}$ is the time for automated auditing.

2 Research Question

Primary RQ:

Given the limitations of manual and tool-siloed auditing in **Rwandan** hybrid cloud environments, including high cost, excessive time, and fragmented oversight [1, 2, 9], can a localized AI-augmented compliance auditor (focused on NIST SP 800-53 and **Rwanda’s NCSA Minimum Cybersecurity Standards (2023)**, excluding auto-remediation) achieve $> 93\%$ classification accuracy and $\geq 50\%$ audit cycle reduction while maintaining transparency and fairness [10–12]?

Secondary RQs:

- a. Does fine-tuned BERT improve log-to-control mapping accuracy compared to baseline classifiers [10]?
- b. Do One-Class SVM/LSTM anomaly detectors and Random Forest/Gradient Boosting models materially enhance detection and prioritization of compliance findings [11]?
- c. Can a training strategy that combines synthetic and public datasets produce valid results for the **Rwandan** environment without introducing unacceptable bias [1, 13]?
- d. What measurable reductions in audit time and effort are observed relative to manual methods, and how does this relate to the broader economic impact of breaches in **Rwanda** [2, 4]?

3 Annotated Literature Survey

3.1 AI-Assisted Compliance and Risk Scoring

Transformer-based Models for Automated Security Audit Classification (Green & Brown, 2023) – This work fine-tuned a BERT model to classify security audit logs and achieved high accuracy ($> 93\%$) on benchmark datasets. The study demonstrates the suitability of transformer architectures for mapping unstructured log messages to compliance controls [5].

Machine Learning for Automated Compliance Auditing: A Deep Learning Approach (Smith et al., 2022) – The authors used LSTM auto-encoders and Random Forest models for anomaly detection and risk scoring. They reported a 50% reduction in audit cycle time and noted that explainable ML (SHAP/LIME) improves trust in automated decisions [6].

AI-Enhanced Whistle-blower Systems and Predictive Compliance Analytics (CentralEyes, 2023) – Describes tools that use AI to analyse historical compliance data and forecast trends. These systems monitor control performance and detect anomalies early, demonstrating the potential for predictive compliance analytics to reduce risk exposure [14].

3.2 Challenges of AI in Compliance Auditing

Expanded Attack Surface – AI/ML systems increase the attack surface; adversarial examples, model theft and data poisoning can undermine compliance results [6]. This underscores the need for secure-by-design AI pipelines.

Data Scarcity and Bias – Quality labelled datasets for **specific African regulatory contexts like Rwanda’s** are scarce. Using synthetic data and public datasets mitigates this but introduces potential bias. Explainable models (SHAP, LIME) are needed to analyse bias and build trust.

Explainability and Ethics – Explainability is crucial when AI recommends compliance decisions. Tools like SHAP can highlight feature contributions to risk scores, ensuring decisions align with legal and ethical requirements [6].

4 Dataset Strategy

4.1 Synthetic Compliance Event Dataset

Because no publicly available dataset exists for Rwanda’s NCSA Minimum Cybersecurity Standards baseline, this research project will construct a synthetic dataset based on the controls defined in NIST SP 800-53 and Rwanda’s Minimum Cybersecurity Standards. NIST 800-53 provides a comprehensive catalogue of security and privacy controls covering access control, audit logging, incident response and system integrity [15]. The synthetic dataset will include:

- **Control Definitions:** Each record will map to a specific control (e.g., AC-2 Account Management or AU-6 Audit Review and Analysis). For Rwanda’s NCSA standards, controls will be abstracted from publicly available policy summaries (system hardening, access control, vulnerability management and incident response).
- **Event Attributes:** Log fields such as timestamp, user ID, action, resource, status code and compliance outcome (compliant/non-compliant). These will simulate common compliance events.
- **Baseline Requirements:** Rwanda’s Minimum Cybersecurity Standards for Essential Service Providers were introduced by the NCSA in July 2023 and require organisations to perform risk assessments, implement system-integrity and access control measures, and maintain incident-response capabilities. The standards adopt a tiered approach based on criticality and mandate regular reviews [4]. These requirements will inform the synthetic dataset’s control taxonomy.
- **Labels:** Each synthetic event will be labelled with the relevant control and a binary compliance status. For anomaly detection tasks, additional labels (normal vs. anomalous) will be generated.

The synthetic dataset ensures that the prototype can be evaluated against the required frameworks without exposing sensitive data.

4.2 Public Datasets for Model Training and Benchmarking

To develop and benchmark ML models, publicly available cybersecurity datasets will be used:

4.3 Data Augmentation and Pre-processing

- **Log Parsing and Template Extraction:** Apply log parsing algorithms (e.g., Drain) to extract structured log templates from unstructured system logs.
- **Synthetic Event Generation:** Use generative models to create variations of log messages while preserving control semantics, increasing dataset diversity.
- **Balancing Classes:** Use oversampling or cost-sensitive learning to handle class imbalance.
- **Partitioning:** Split the dataset into training (70%), validation (15%) and test (15%) sets.

5 Methods to be Used

5.1 Model Development

The project will train and test several ML models:

Table 1: Public Datasets for Model Development

Dataset	Description	Use in Project	Citation
CICIDS 2017	A benchmark intrusion-detection dataset that includes five days of network traffic with benign and various attack scenarios (DoS, DDoS, brute force, infiltration, web attacks and botnets) [16].	Used to train baseline classifiers and anomaly detectors (Random Forest, SVM, LSTM) before applying them to synthetic compliance data.	UNB CIC [16]
NSL-KDD / UNSW-NB15	Modern network intrusion datasets containing multiple attack types. UNSW-NB15 includes nine attack categories and 49 flow features [17].	Additional dataset for testing cross-dataset generalisation and for augmenting training data.	Public [17]
CERT Insider Threat Dataset	A public dataset capturing user activity logs (logins, emails, file transfers) with labelled insider threat scenarios [18].	Provides realistic user behaviour patterns for generating synthetic compliance events and training anomaly detectors.	CERT [18]
LogHub	A repository of open-source system logs from distributed applications (e.g., Hadoop, OpenStack, Linux) labelled for anomaly detection tasks [19].	Used to train log parsing and message template extraction models, to be adapted to the synthetic compliance dataset.	LogHub [19]

- **NLP for Log Mapping:** Fine-tune a pre-trained BERT model for log-to-control mapping [10].
- **Anomaly Detection:** Implement One-Class SVM and LSTM autoencoders for detecting unusual activities [11].
- **Risk Scoring:** Use Random Forest and Gradient Boosting models to prioritize compliance findings based on severity and impact.

5.2 Risk Scoring Model

The risk score for each compliance finding is calculated using a weighted ensemble:

$$R_{\text{total}} = \alpha \cdot R_{\text{severity}} + \beta \cdot R_{\text{likelihood}} + \gamma \cdot R_{\text{business impact}} \quad (3)$$

where $\alpha + \beta + \gamma = 1$ and each risk component is normalized to $[0, 1]$.

5.3 BERT Fine-tuning Mathematical Foundation

The BERT model fine-tuning for compliance log classification involves optimizing the cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(p_{ij}) \quad (4)$$

where N is the number of samples, C is the number of compliance categories, y_{ij} is the true label, and p_{ij} is the predicted probability.

5.4 Anomaly Detection Formulation

For One-Class SVM, the optimization objective is to find a hyperplane that separates the data from the origin:

$$\min_{w, \xi, \rho} \frac{1}{2} \|w\|^2 + \frac{1}{\nu n} \sum_{i=1}^n \xi_i - \rho \quad (5)$$

subject to $(w \cdot \phi(x_i)) \geq \rho - \xi_i$ and $\xi_i \geq 0$.

5.5 System Integration and Evaluation

System Integration: Deploy the prototype on a Kubernetes testbed for realistic simulation.

Evaluation: Measure against defined metrics using statistical validation.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Validation: Compare automated auditing results against manual processes; assess limitations in dataset bias and model transparency.

5.6 Performance Evaluation Metrics

The system will be evaluated using multiple metrics:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (9)$$

Additionally, the Area Under the ROC Curve (AUC-ROC) will be calculated as:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt \quad (10)$$

where TPR is True Positive Rate and FPR is False Positive Rate.

6 Anticipated Results

The anticipated results are:

- A working prototype auditor localized for Rwanda’s NCSA Minimum Cybersecurity Standards (2023) and NIST 800-53, demonstrating accuracy > 93% [5,6] and audit cycle time reduction > 50% [7].
- Insights into dataset availability, explainability, and bias mitigation in the **Rwandan** hybrid cloud context [8,20].
- A peer-reviewed quality final report and defense presentation documenting findings.
- Empirical validation of the cost-benefit equation:

$$\text{Cost Savings} = (T_{\text{manual}} \times C_{\text{hourly}}) - (T_{\text{automated}} \times C_{\text{hourly}} + C_{\text{system}}) \quad (11)$$

where C_{hourly} is the hourly cost of manual auditing and C_{system} is the system operational cost.

7 Significance

This research project contributes to both academic knowledge and practical impact. By developing the first AI compliance auditing prototype specifically for the **Rwandan context**, it strengthens the nation’s capacity to address local cybersecurity needs and enhance digital resilience. Furthermore, it offers a **scalable foundation and replicable model** for future industrial adoption within Rwanda and for other African nations as they develop their own specific cybersecurity standards. The economic impact can be quantified as:

$$\text{ROI} = \frac{\text{Annual Savings} - \text{Implementation Cost}}{\text{Implementation Cost}} \times 100\% \quad (12)$$

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